

Mirembe Peace Mercy

Computer Scientist | Aspiring Embedded Systems Engineer
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PROFILE

Computer scientist specialising in embedded systems, robotics, AR/VR, and full-stack development, with a focus on solving real-world challenges in energy, digital access, and immersive experiences. Passionate about putting Uganda at the frontier of innovation through hands-on prototypes, open research, and community-driven technical education.

EDUCATION

Bachelor of Science in Computer Science

Uganda Christian University | May 2024 – Present

Specialisation: Embedded Systems & Robotics

Relevant Electives (Year 2):

- Digital Electronics
- Embedded Systems and Microcontrollers
- Robotics and Control Systems

WORK EXPERIENCE

Software Developer

Pearl Rhythm Foundation | Feb 2025 – Aug 2025

- Built a fully immersive, interactive web experience of a museum showcasing musical artifacts of indigenous tribes
- Incorporated spatial sound, text, and video demonstrations of cultural artifacts
- **Live project:** virtual.pearlrhythmfoundation.org

Software Developer and Board Member

CEMAK Technologies Limited - Mukono Uganda | March 2026 – Till date

- Design and develop full-stack web applications for clients across multiple industries
- Contribute to strategic decision-making as a member of the Board of Directors
- Participate in business development, project planning, and technology direction

PERSONAL PROJECTS

Kawewo (AC 3000) – Smart Cooling System v2

Personal Project

Second prototype of an energy-efficient smart cooling system targeting real-world energy challenges in Uganda.

- Designed and prototyped the system using ESP32 microcontroller
- Sensors: DHT11 (temperature), PIR (occupancy detection), voltage sensor
- Implemented adaptive fan speed control based on temperature thresholds
- Power-saving mode via occupancy detection logic
- Built real-time telemetry system using WebSockets
- Developed a React + TypeScript dashboard for live monitoring
- Proposed ML optimisation model targeting 25–40% reduction in energy consumption

AshabaMusic.com Platform – Full Stack

Personal Project

A music distribution and monetisation platform for African artists.

- Integrated Pesapal API for mobile money and card payments
- Backend built with NestJS + PostgreSQL + TypeORM
- Secure payment workflow with webhook verification
- Features: music sales (albums, singles), artist booking, and fan tipping system
- Deployed via Railway (backend) and Netlify (frontend)

Typing Game – Educational Web App

Team Project

- Vue + Vite web game to improve computing students' typing speed

Digital Scoreboard – IoT Hardware Project

Team Project

- LED matrix scoreboard with remote WiFi control via mobile and web interface for campus sports events

RESEARCH EXPERIENCE

Geothermal Cooling Research – GSHP Systems

Uganda Christian University | Ongoing

- Conducting research on Ground Source Heat Pump (GSHP) systems suitable for Uganda's climate
- Focus: energy efficiency in office buildings and cost-benefit analysis vs. traditional air conditioning
- Proposed sustainable cooling solutions for low-resource environments
- Target: 30–50% reduction in building energy consumption

LEADERSHIP & EXPERIENCE

President – Gaming & AR Club (GEAR Chapter)

Uganda Christian University | 2024 – Present

- Lead a student club focused on gaming, AR/VR, and immersive technologies
- Organise and teach sessions on Unity (C#) game development and WebXR / Three.js
- Spearhead development of interactive projects for university exhibitions and launches

Academic Leader – Computing Students Association

Uganda Christian University

- Coordinate academic support activities for computing students
- Promote project-based learning and peer education initiatives

Residential Assistant – UCU halls of residence

Uganda Christian University

- Coordinate accommodation support for student residents

TECHNICAL SKILLS

Programming & Frameworks: JavaScript, TypeScript, C, C++, Python, SQL

Frontend: React, Three.js, Vue + Vite, Next.js, Tailwind CSS, WebXR

Backend: Node.js, NestJS, PostgreSQL, TypeORM, MySQL, MongoDB

Hardware & Embedded: ESP32, Arduino, DHT11 / PIR / Voltage sensors, PWM control

Tools & Platforms: Git / GitHub, Blender, Figma, VSCode, Railway, Netlify

Emerging Tech: WebXR, AR/VR (Unity C#), Machine Learning (concept-level)

KEY ACHIEVEMENTS

- Built multi-platform VR/AR experiences accessible on mobile, desktop, and VR headsets
- Designed smart cooling prototype addressing real-world energy inefficiency challenges in Uganda
- Led and taught technical workshops in game development and immersive technologies
- Developed a full-stack, payment-enabled music distribution platform
- Conducting active research on geothermal cooling solutions for Sub-Saharan Africa

LANGUAGES

English: Full professional proficiency

Luganda: Native